WEAK APPROXIMATION FOR CAMPANA POINTS ON PROJECTIVE SPACES

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Abstract. We prove weak approximation for Campana points on the projective line with log Fano boundary, and on projective n-space with either $n+3$ hyperplanes in general position or two transverse smooth quadrics, each of coefficient $1/2$.

1. Introduction

The paper is devoted to weak approximation for Campana points on projective spaces over number fields. Let X be a smooth projective variety over a number field k, and let

$$D = \sum_i \left(1 - \frac{1}{n_i}\right) D_i, \qquad 2 \le n_i < \infty,$$

be an orbifold divisor. Fix a finite set S of places containing the archimedean places and a proper flat integral model $\mathscr{X}$ of X over $\mathcal{O}_S$. A point $x \in (X \setminus \operatorname{Supp} D)(k)$ is an S-Campana point if

$$n_v(\mathscr{D}_{i,\lambda}, x) = 0 \quad \text{or} \quad n_v(\mathscr{D}_{i,\lambda}, x) \ge n_i \qquad (v \notin S,\ \lambda \in I_i(v)),$$

where $\mathscr{D}_i$ is the closure of D_i in $\mathscr{X}$ and $\mathscr{D}_{i,\lambda}$ runs through the irreducible components of $\mathscr{D}_i \times_{\mathcal{O}_S} \mathcal{O}_v$. Here n_v is the local intersection multiplicity defined by the colength of the pulled-back ideal. We use [NS24, Definitions 2.3–2.4] throughout. Campana weak approximation means that these points are dense in the product of the local Campana spaces. These componentwise inequalities differ from the divisibility conditions for Darmon points; see [PSTVA21, §3.2] and [MNS24, §2].

Campana introduced orbifolds to retain the contribution of multiple fibres to the geometry and arithmetic of a fibration [Cam04, Cam05]. A smooth orbifold is special if, on suitable smooth birational models, it has no orbifold fibration onto a positive-dimensional base of general type [Cam11, §4.3]. The following conjecture relates this property to arithmetic.

Conjecture (Campana, [Cam11, Conjecture 12.21(1)])**.** *Let (X, D) be a smooth projective Campana orbifold over a number field k, with simple normal crossings boundary and finite integral orbifold multiplicities. Then $(X_{\overline{k}}, D_{\overline{k}})$ is special if and only if there exist a finite extension k'/k, a finite set $S' \supseteq \Omega_{k'}^{\infty}$, and a proper integral model over $\mathcal{O}_{S'}$ with Zariski-dense S'-Campana points in $X_{k'}$.*

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Smooth log Fano orbifolds are special [Cam11, Corollary 6.9]. Abramovich gives a finer formulation for a constellation supported by a firmament, whose integral points are predicted to be potentially dense precisely when the constellation is special [Abr09, Conjecture 2.4.19]. We use the componentwise Campana conditions, not those of an arbitrary firmament. The potential-density conjecture allows a field extension and enlargement of S; the approximation statements below hold over k for every finite set S and every proper flat integral model over $\mathcal{O}_S$.

Nakahara and Streeter prove Campana weak approximation on $\mathbf{P}^n$ with at most $n+1$ hyperplanes in general position and arbitrary finite multiplicities, and with $n+2$ hyperplanes of multiplicity two [NS24, Theorem 1.3 and Corollary 1.4]. They also relate Campana weak weak approximation to the Hilbert property [NS24, Theorem 1.1]. Mitankin, Nakahara and Streeter develop a semi-integral Brauer–Manin obstruction and prove weak approximation for a single smooth quadric boundary with arbitrary finite multiplicity [MNS24, Theorem 1.3(i)]. Related counting problems are studied in [PSTVA21].

Our first result concerns the projective line.

Theorem 1.1. *Let P_i be distinct closed points of $\mathbf{P}^1_k$, and let*

$$D = \sum_{i\in I} \left(1 - \frac{1}{n_i}\right) P_i, \qquad 2 \le n_i < \infty.$$

Suppose that

$$\sum_{i\in I} \left(1 - \frac{1}{n_i}\right) [k(P_i) : k] < 2. \tag{1.1}$$

Then $(\mathbf{P}^1_k, D)$ satisfies Campana weak approximation for every finite set $S \supseteq \Omega^\infty_k$ and every proper flat integral $\mathcal{O}_S$-model of $\mathbf{P}^1_k$. More precisely, global S-Campana points outside $\operatorname{Supp} D$ *are dense in the product of the corresponding local Campana spaces.*

The degree inequality is equivalent to the ampleness of $2H - D$, where H is a rational point of $\mathbf{P}^1$. Our next two results concern higher-dimensional projective spaces.

Theorem 1.2. *Let $n \ge 2$, and let $E_1, \dots, E_{n+3} \subset \mathbf{P}^n_k$ be distinct k-hyperplanes in general position. Set*

$$D = \frac{1}{2}\sum_{i=1}^{n+3} E_i, \qquad U = \mathbf{P}^n_k \setminus \bigcup_{i=1}^{n+3} E_i.$$

For every finite set $S \supseteq \Omega^\infty_k$ and every proper flat integral $\mathcal{O}_S$-model of $\mathbf{P}^n_k$, the S-Campana points in $U(k)$ are dense in the product of the corresponding local Campana spaces. In particular, $(\mathbf{P}^n_k, D)$ satisfies Campana weak approximation.

Here general position means that any $n+1$ defining linear forms are linearly independent. For $n = 2$ this gives five lines with no triple intersection.

Theorem 1.3. *Let $n \ge 2$, and let $C_1, C_2 \subset \mathbf{P}^n_k$ be smooth k-quadric hypersurfaces meeting transversely. Set*

$$D = \frac{1}{2}(C_1 + C_2), \qquad U = \mathbf{P}^n_k \setminus (C_1 \cup C_2).$$

For every finite set $S \supseteq \Omega_k^\infty$ and every proper flat integral $\mathcal{O}_S$-model of $\mathbf{P}^n_k$, the S-Campana points in $U(k)$ are dense in the product of the corresponding local Campana spaces. In particular, $(\mathbf{P}^n_k, D)$ satisfies Campana weak approximation.

Transversality means that $C_1 \cap C_2$ is smooth of codimension two; neither the quadrics nor their intersection need have rational points. The pairs in Theorems 1.2 and 1.3 are log Fano, since $-(K_{\mathbf{P}^n} + D)$ is respectively equivalent to $\frac{n-1}{2}H$ and $(n-1)H$.

The proofs first choose a rational point on the base, then a finite cover with a rational point above it. In the higher-dimensional cases, strong approximation away from a codimension-two subset ensures that the chosen point reduces onto at most one boundary component outside the prescribed approximation places. For Theorem 1.1, spherical uniformization gives a rational curve. For the other two theorems, the covers are smooth intersections of two quadrics. In dimension at least three, they satisfy weak approximation. For the surface cases, we either arrange that the Brauer group is constant or preserve Brauer evaluations under local modifications. The essential local arguments treat every new bad place of the cover. Section 2 records the preliminaries, and Sections 3–5 prove the three theorems in order.

Terminology. Write $\mathcal{O}_S$ for the ring of S-integers in k; the same notation with S' refers to the corresponding ring in the specified extension field. Write Ω_k for the places of k, Ω_k^∞ for the archimedean places, k_v for the completion, and $\mathbf{A}_k$ for the adeles. For finite v, write $\mathcal{O}_v$, $\mathbf{F}_v$, ϖ_v and ord_v for its valuation ring, residue field, a uniformizer and the normalized valuation. We use $\mathrm{Br}(Y) = H^2_{\text{ét}}(Y, \mathbf{G}_{\mathrm{m}})$ and $\mathrm{Br}_0(Y) = \mathrm{Im}(\mathrm{Br}(k) \to \mathrm{Br}(Y))$. For proper Y, the notation $Y(\mathbf{A}_k)^{\mathrm{Br}(Y)}$ denotes the adelic points orthogonal to $\mathrm{Br}(Y)$ under the Brauer–Manin pairing.

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2. Preliminaries

Let X be smooth, projective and geometrically integral, and let $D = \sum_i (1 - 1/n_i)D_i$ with distinct prime divisors D_i defined over k and integers $2 \le n_i < \infty$. The D_i need not be geometrically irreducible. Put $X^\circ = X \setminus \operatorname{Supp} D$, and fix a proper flat integral model $\mathscr{X}$ over $\mathcal{O}_S$, with boundary closures $\mathscr{D}_i$.

For $v \notin S$, write

$$D_i \times_k k_v = \bigcup_{\lambda \in I_i(v)} D_{i,\lambda},$$

where the $D_{i,\lambda}$ are the k_v-irreducible components, and let $\mathscr{D}_{i,\lambda}$ be their reduced closures in $\mathscr{X}_{\mathcal{O}_v}$. These are the irreducible components of $\mathscr{D}_i \times_{\mathcal{O}_S} \mathcal{O}_v$ used in [NS24, Definition 2.2]; they are not the irreducible components of the special fibre.

A point $x \in X(k_v)$ extends uniquely to a section $\widetilde{x} : \operatorname{Spec} \mathcal{O}_v \to \mathscr{X}_{\mathcal{O}_v}$. For a closed subscheme $\mathscr{E} \subset \mathscr{X}_{\mathcal{O}_v}$ with ideal sheaf $\mathcal{I}_{\mathscr{E}}$, put $J_{\mathscr{E},x} = \widetilde{x}^{-1}\mathcal{I}_{\mathscr{E}} \cdot \mathcal{O}_v$ and define its *local*

intersection multiplicity with x by

$$n_v(\mathscr{E},x)=\begin{cases}\operatorname{length}_{\mathcal{O}_v}(\mathcal{O}_v/J_{\mathscr{E},x}), & J_{\mathscr{E},x}\neq 0,\\ \infty, & J_{\mathscr{E},x}=0.\end{cases}$$

Equivalently, $J_{\mathscr{E},x}=\mathfrak{m}_v^{n_v(\mathscr{E},x)}$ when the intersection multiplicity is finite. If $\mathscr{E}$ is locally cut out by one equation f near the reduction of x, then $n_v(\mathscr{E},x)=\operatorname{ord}_v(f(x))$, with $\operatorname{ord}_v(0)=\infty$. Applied to each $\mathscr{D}_{i,\lambda}$, this is the intersection multiplicity of [NS24, Definition 2.3].

We display the chosen model in the notation for Campana points. The second argument D denotes the divisor on the generic fibre; all boundary closures are taken in $\mathscr{X}$. Following [NS24, Definition 2.4], set

$$\begin{aligned}&\mathcal{C}_v(\mathscr{X},D)=\{x\in X(k_v): n_v(\mathscr{D}_{i,\lambda},x)\in\{0\}\cup\mathbf{Z}_{\geq n_i}\cup\{\infty\}\text{ for all } i,\lambda\in I_i(v)\}, && v\notin S,\\ &\mathcal{C}_v(\mathscr{X},D)=X(k_v), && v\in S,\\ &\mathcal{C}_S(\mathscr{X},D)=\{x\in X^\circ(k): x\in\mathcal{C}_v(\mathscr{X},D)\text{ for every } v\notin S\}.\end{aligned}$$

Thus the inequalities are imposed separately on the local irreducible components, not merely on the total intersection multiplicity with a k-prime divisor. If D_i is geometrically integral, $I_i(v)$ has one element, and we abbreviate the corresponding intersection multiplicity to $n_v(\mathscr{D}_i,x)$. Campana weak approximation is the density of $\mathcal{C}_S(\mathscr{X},D)$ in $\prod_v\mathcal{C}_v(\mathscr{X},D)$ with the product topology. We require the global approximating point to lie outside the boundary, although local points on the boundary are allowed. Only finitely many local approximations are prescribed, but the global point must satisfy the inequalities at every finite $v\notin S$.

Lemma 2.1. *The space $\mathcal{C}_v(\mathscr{X},D)$ is open in $X(k_v)$, and $X^\circ(k_v)\cap\mathcal{C}_v(\mathscr{X},D)$ is dense in $\mathcal{C}_v(\mathscr{X},D)$.*

Proof. At a finite place, reduction of sections modulo a fixed power of $\mathfrak{m}_v$ is locally constant for the v-adic topology. Near the reduction of a section, choose finitely many generators of the ideal of each $\mathscr{D}_{i,\lambda}$. Its intersection multiplicity is the minimum of their valuations. A finite minimum is locally constant; near a common zero all these valuations exceed any fixed bound. There are only finitely many local components, and all weights are finite, so the Campana conditions are open. The assertion is immediate at places of S. Finally, a proper closed subset of a smooth geometrically integral k_v-variety contains no nonempty v-adic open subset. Applying this to $\operatorname{Supp} D$ proves the density assertion. □

By [NS24, Lemma 2.8], to prove Theorems 1.1–1.3 for arbitrary proper flat integral models, it suffices to treat the standard models $\mathbf{P}^d_{\mathcal{O}_S}$ for every finite $S\supseteq\Omega_k^\infty$. Indeed, any such model, together with its boundary closures, becomes standard after enlarging S, while Lemma 2.1 allows the original Campana conditions at the added places to be imposed as open local requirements.

In the proofs we therefore fix a finite $S\supseteq\Omega_k^\infty$ and work with the standard models $\mathbf{P}^1_{\mathcal{O}_S}$ and $\mathbf{P}^n_{\mathcal{O}_S}$, with all boundary closures taken in the corresponding model. The auxiliary sets $T,T'\supseteq S$ do not replace S in the Campana conditions.

Avoiding a codimension-two subset. We use the standard fact that $\mathbf{P}^n_k \setminus Z$ satisfies strong approximation whenever $Z \subset \mathbf{P}^n_k$ is closed of codimension at least two; see, for example, [Che22, Proposition 2.7]. Let $\mathscr{Z}$ be a closed model of Z and put $\mathcal{W} = \mathbf{P}^n_{\mathcal{O}_S} \setminus \mathscr{Z}$. For any finite $T \supseteq S$ such that $\mathcal{W}(\mathcal{O}_v) \neq \varnothing$ for every $v \notin T$, and any nonempty open sets $V_v \subset (\mathbf{P}^n \setminus Z)(k_v)$, $v \in T$, strong approximation gives a rational point a belonging to every V_v and satisfying

$$a \in \mathcal{W}(\mathcal{O}_v) \qquad (v \notin T).$$

Thus the reduction of a avoids $\mathscr{Z}$ outside T; no such reduction condition is imposed at the prescribed approximation places.

We shall use the following arithmetic of intersections of two quadrics.

Proposition 2.2. *Let $Y \subset \mathbf{P}^{m+2}_k$ be a smooth geometrically integral intersection of two quadrics, with $m \geq 2$ and $P_0 \in Y(k)$. Then $\overline{Y(k)} = Y(\mathbf{A}_k)^{\mathrm{Br}(Y)}$, where closure is taken in the usual adelic topology (the product topology since Y is proper). If $m = 2$, the group $\mathrm{Br}(Y)/\mathrm{Br}(k)$ is finite and killed by 2. If $m \geq 3$, then $\mathrm{Br}(Y) = \mathrm{Br}(k)$ and Y satisfies ordinary weak approximation.*

Proof. If $m = 2$, then Y is a del Pezzo surface of degree 4. The density statement is [SS91, Theorem 0.1], and $\mathrm{Br}(Y)/\mathrm{Br}(k)$ is isomorphic to 0, $\mathbf{Z}/2$ or $(\mathbf{Z}/2)^2$ by [VAV14, Remark 1.2(1)].

For $m = 3$, the Brauer–Manin approximation statement follows from [CTS92, Theorem 1]; for $m \geq 4$, ordinary weak approximation is just [CTSSD87, Theorem 3.11]. We explain why the former statement also gives ordinary weak approximation.

For every $m \geq 3$, the Lefschetz hyperplane theorem gives $\mathrm{Pic}(Y_{\overline{k}}) = \mathbf{Z}[\mathcal{O}_Y(1)]$ and $H^2_{\text{ét}}(Y_{\overline{k}}, \mu_r) = \mathbf{Z}/r$, generated by the hyperplane class. In the Kummer sequence

$$0 \longrightarrow \mathrm{Pic}(Y_{\overline{k}})/r \longrightarrow H^2_{\text{ét}}(Y_{\overline{k}}, \mu_r) \longrightarrow \mathrm{Br}(Y_{\overline{k}})[r] \longrightarrow 0,$$

the first map is therefore an isomorphism. Thus $\mathrm{Br}(Y_{\overline{k}}) = 0$. The hyperplane class is defined over k, so Hochschild–Serre gives

$$\mathrm{Br}(Y)/\mathrm{Br}(k) \hookrightarrow H^1(k, \mathrm{Pic}(Y_{\overline{k}})) = H^1(k, \mathbf{Z}) = 0.$$

Hence $\mathrm{Br}(Y) = \mathrm{Br}(k)$, and global reciprocity gives $Y(\mathbf{A}_k)^{\mathrm{Br}(Y)} = Y(\mathbf{A}_k)$. □

The rational point in the proposition is part of its hypotheses; it will be supplied by the construction of each cover. In the surface case, the result of Salberger–Skorobogatov is being applied only after this point is available. The distinction between surfaces and higher-dimensional varieties explains the extra discriminant argument in Section 4 and the use of the following residue lemma in the two-conic case.

Lemma 2.3. *Let R be the valuation ring of a nonarchimedean local field K of odd residue characteristic, with residue field $\mathbf{F}$. Let $\mathcal{M}$ be smooth, separated and of finite type over R. If $P, Q \in \mathcal{M}(R)$ reduce to a locally closed subvariety $A \simeq \mathbf{A}^r_{\mathbf{F}}$ of $\mathcal{M}_{\mathbf{F}}$, then $\beta(P) = \beta(Q)$ for every $\beta \in \mathrm{Br}(\mathcal{M}_K)[2]$.*

Proof. By [Bri15, Corollary 2.5 and Proposition 5.1], we have

$$\partial : \mathrm{Br}(\mathcal{M}_K)[2] \longrightarrow H^1(\mathcal{M}_{\mathbf{F}}, \mathbf{Z}/2), \qquad \partial_K(\beta(P)) = \bar{P}^*(\partial\beta).$$

The smoothness of $\mathcal{M}$ over R makes its special fibre a regular divisor; if it is disconnected, the residue map is taken on each component. The local residue map is an isomorphism $\partial_K : \mathrm{Br}(K)[2] \xrightarrow{\sim} H^1(\mathbf{F}, \mathbf{Z}/2)$, since R is henselian and $\mathrm{Br}(\mathbf{F}) = 0$ for a finite field. By Kummer theory, $H^1(\mathbf{A}^r_{\mathbf{F}}, \mathbf{Z}/2) = H^1(\mathbf{F}, \mathbf{Z}/2)$, since the only units of $\mathbf{F}[t_1, \ldots, t_r]$ are constants and its Picard group is zero. Thus $(\partial\beta)|_A$ is constant, and its evaluations at $\bar{P}$ and $\bar{Q}$ agree. Injectivity of ∂_K then gives $\beta(P) = \beta(Q)$ in $\mathrm{Br}(K)$. □

The reductions $\bar{P}$ and $\bar{Q}$ need not be the same point. The lemma compares their evaluations through the residue class on A, rather than through local constancy in a neighbourhood of one K-point. It does not require β to extend to $\mathrm{Br}(\mathcal{M})$: its possible ramification along the special fibre is precisely what the residue map records.

3. The projective line

Throughout this section, all boundary closures are taken in the standard model $\mathbf{P}^1_{\mathcal{O}_S}$.

3.1. **Geometric signatures.** Assume the hypotheses of Theorem 1.1 and $D \neq 0$. Let r be the number of points of $\mathrm{Supp}(D_{\bar{k}})$. Each geometric boundary point contributes at least $1/2$, so (1.1) implies $r \leq 3$. If there are three, their ordered multiplicities $a \leq b \leq c$ satisfy $1/a + 1/b + 1/c > 1$. Thus $a = 2$, and either $b = 2$, or $b = 3$ and $c < 6$. The possible signatures are

$$(2, 2, m)\ (m \geq 2), \qquad (2, 3, 3), \qquad (2, 3, 4), \qquad (2, 3, 5). \tag{3.1}$$

To include $r = 1$ and $r = 2$ in a single construction, choose an auxiliary divisor $\Delta \geq D$ as follows. If $r = 1$, the point is rational; add another rational point of the same multiplicity. If $r = 2$, replace both multiplicities by their maximum N. If $r = 3$, take $\Delta = D$. These operations are Galois invariant and give

$$\Delta = \sum_{j \in J} \left(1 - \frac{1}{e_j}\right) Q_j, \qquad Z = \mathrm{Supp}\,\Delta, \qquad U_\Delta = \mathbf{P}^1 \setminus Z. \tag{3.2}$$

Every original P_i is a Q_j with $e_j \geq n_i$. The geometric signature of Δ is (N, N) or one of (3.1).

The auxiliary divisor is necessary for the uniformization used below. An orbifold with one nontrivial point, or two points of unequal multiplicities, does not have a scheme as its universal orbifold cover; see [BN06, Propositions 5.5 and 5.7]. We uniformize $(\mathbf{P}^1, \Delta)$, not the original pair in these cases. This neither replaces the local conditions for D by those for Δ nor requires n_i to divide e_j.

3.2. **Pointed descent and the deck group.** The spherical signatures in (3.2) admit universal orbifold covers over $\mathbf{P}^1_{\overline{k}}$; see Behrend–Noohi [BN06, Proposition 5.5]. Their geometric deck groups are recorded below.

Signature of $\Delta_{\overline{k}}$	H	$\deg \bar{\pi} = \lvert H\rvert$
(N, N)	$\mathbf{Z}/N\mathbf{Z}$	N
$(2, 2, m)$	D_m	$2m$
$(2, 3, 3)$	A_4	12
$(2, 3, 4)$	S_4	24
$(2, 3, 5)$	A_5	60

Here D_m has presentation $\langle r, s : r^m = s^2 = 1,\ srs = r^{-1}\rangle$; in particular, $D_2 \simeq (\mathbf{Z}/2)^2$. For geometric multiplicities $\varepsilon_1, \dots, \varepsilon_r$, the orbifold presentation is

$$H \simeq \langle \gamma_1, \dots, \gamma_r : \gamma_i^{\varepsilon_i} = 1,\ \gamma_1 \cdots \gamma_r = 1\rangle.$$

The stabilizer of a geometric point above Q_j is cyclic of order e_j. The action is free precisely away from the ramification divisor. The table specifies the geometric group, not a constant group scheme over k.

Proposition 3.1. *There is a geometrically Galois finite morphism $\pi : Y \to \mathbf{P}^1_k$, with $Y \simeq \mathbf{P}^1_k$, branched precisely over Z. Its deck group scheme*

$$G = \underline{\mathrm{Aut}}_{\mathbf{P}^1_k}(Y)$$

is finite étale over k, with $G(\overline{k}) = H$. If $R_j = (\pi^{-1}(Q_j))_{\mathrm{red}}$ and $Y_U = \pi^{-1}(U_\Delta)$, then

$$\pi^* Q_j = e_j R_j, \tag{3.3}$$

and $Y_U \to U_\Delta$ is a left G-torsor.

Proof. Choose $b \in U_\Delta(k)$ and $y_b \in \bar{\pi}^{-1}(b)(\overline{k})$. Write $\Gamma_k = \mathrm{Gal}(\overline{k}/k)$. For $\sigma \in \Gamma_k$, the conjugate cover is again universal for the same orbifold, since Δ is defined over k. Uniqueness of the universal cover, and the simply transitive action of H on the fibre above b, give a unique isomorphism of pointed covers

$$\phi_\sigma : {}^\sigma(\bar{Y}, \bar{\pi}, y_b) \xrightarrow{\sim} (\bar{Y}, \bar{\pi}, y_b).$$

Consequently $\phi_{\sigma\tau} = \phi_\sigma \circ {}^\sigma\phi_\tau$. The cover and the point are defined over a finite extension. Taking a finite Galois extension containing their fields of definition, uniqueness makes these maps a finite descent datum. Effective descent yields $\pi : Y \to \mathbf{P}^1_k$ and $y_b \in Y(k)$. Since Y is smooth projective of geometric genus zero, $Y \simeq \mathbf{P}^1_k$.

The induced action on H is

$$\rho_\sigma(h) = \phi_\sigma \circ {}^\sigma h \circ \phi_\sigma^{-1} \qquad (h \in H). \tag{3.4}$$

The cocycle identity for ϕ shows that ρ is a continuous Γ_k-action by group automorphisms. Finite groups with such action correspond to finite étale k-group schemes [Stacks, Tag 04JI]. Let G correspond to (H, ρ). The action on $\bar{Y}$ descends with it. Equivalently, for a k-scheme T, one has $G(T) = \mathrm{Aut}_{\mathbf{P}^1_T}(Y_T)$; this identifies G with the group scheme in the statement.

Over U_Δ the cover is finite étale. After extension to $\overline{k}$, H acts simply transitively on each geometric fibre. Thus the natural morphism

$$G \times_k Y_U \longrightarrow Y_U \times_{U_\Delta} Y_U, \qquad (g, y) \longmapsto (g \cdot y, y), \tag{3.5}$$

is an isomorphism, which proves the torsor assertion. Finally, the geometric ramification indices give (3.3). □

In particular, $G(k) = H^{\Gamma_k}$ is the group of deck transformations defined over k, and need not have $|H|$ elements. The group scheme G is constant exactly when ρ is trivial. Geometric Galoisness therefore does not assert that $k(Y)/k(t)$ is Galois.

3.3. **Twists and local specialization.** Let σ_Y denote the semilinear Galois action on $Y_{\overline{k}}$ coming from Y/k; then $\rho_\sigma(h) = \sigma_Y h \sigma_Y^{-1}$. We fix the twisting convention. A continuous 1-cocycle for G is a family $c_\sigma \in H$ satisfying

$$c_{\sigma\tau} = c_\sigma \rho_\sigma(c_\tau). \tag{3.6}$$

It defines the twist Y_c by the semilinear action

$$\sigma \star y = c_\sigma \cdot \sigma_Y(y). \tag{3.7}$$

As deck transformations fix the target, the map descends to $\pi_c : Y_c \to \mathbf{P}^1_k$. The descended deck group is the inner form G_c, with Galois action on H given by

$$\rho^c_\sigma(h) = c_\sigma \rho_\sigma(h) c_\sigma^{-1}. \tag{3.8}$$

Thus $(Y_c)_{U_\Delta} \to U_\Delta$ is a G_c-torsor, not in general a torsor under the unchanged G; see [San26, §2.2]. If G is commutative, this inner form is G itself.

For $a \in U_\Delta(k)$, choose $y_a \in \pi^{-1}(a)(\overline{k})$ and let c_σ be the unique element of H with

$$c_\sigma \cdot \sigma_Y(y_a) = y_a. \tag{3.9}$$

Freeness on the fibre gives (3.6). The resulting pointed twist is

$$\pi_a : Y_a \to \mathbf{P}^1_k, \qquad y_a \in Y_a(k), \qquad \pi_a(y_a) = a. \tag{3.10}$$

Since Y_a has geometric genus zero and a rational point, $Y_a \simeq \mathbf{P}^1_k$.

Put $R = \bigcup_j R_j$. Let $T \supseteq S$ be a finite set containing the places dividing $|H|$ such that the following data extend over $\mathcal{O}_T$: π to a finite morphism $\mathscr{Y} \to \mathbf{P}^1_{\mathcal{O}_T}$ with $\mathscr{Y}$ smooth and proper; G to a finite étale group scheme $\mathcal{G}$ acting on $\mathscr{Y}$ over the base; and Z, R to finite étale divisors $\mathscr{Z}, \mathscr{R}$ with distinct geometric points in each fibre. Require also

$$\pi^* \mathscr{Q}_j = e_j \mathscr{R}_j. \tag{3.11}$$

These are spreading-out conditions on the original cover, chosen before any a or twist. In particular, geometric deck transformations extend over the integral closure of each valuation ring at $v \notin T$.

For $v \notin T$, the irreducible components $\mathscr{Q}_{j,\lambda}$ of $\mathscr{Z}_{\mathcal{O}_v}$ are pairwise disjoint and finite étale over $\mathcal{O}_v$. A section of $\mathbf{P}^1_{\mathcal{O}_v}$ can meet at most one of them. If it meets $\mathscr{Q}_{j,\lambda}$, its reduction supplies an $\mathbf{F}_v$-point of this component. Since $\mathcal{O}_v$ is henselian, a connected finite étale $\mathcal{O}_v$-scheme with such a point has degree one; hence $\mathscr{Q}_{j,\lambda} \simeq \operatorname{Spec} \mathcal{O}_v$. Near that reduction,

$\mathscr{Q}_j$ and $\mathscr{Q}_{j,\lambda}$ have the same ideal. Consequently, for every local section, at most one local boundary intersection multiplicity is positive, and in that case

$$n_v(\mathscr{Q}_j, x) = n_v(\mathscr{Q}_{j,\lambda}, x), \qquad n_v(\mathscr{Q}_{h,\mu}, x) = 0 \quad ((h,\mu) \neq (j,\lambda)). \tag{3.12}$$

Thus outside T the calculation with $\mathscr{Q}_j$ also verifies each of the local-component inequalities in [NS24, Definition 2.4]. At $T \setminus S$, the original Campana conditions will instead be preserved by local approximation.

Lemma 3.2. *Let $v \notin T$, put $K = k_v$, and let $\pi_c : Y_c \to \mathbf{P}^1_K$ be any twist of π_K. If $q \in Y_c(K)$ and $\pi_c(q)$ reduces into $\mathscr{Z}$, then there is $r \in Y_c(K)$ with $\pi_c(r) \in Z(K)$. Consequently, if $Y_c(K) \neq \varnothing$, there is a nonempty open $W \subset Y_c(K)$ such that $\pi_c(W) \subset \mathcal{C}_v(\mathbf{P}^1_{\mathcal{O}_S}, D) \cap U_\Delta(K)$.*

Proof. Identify $(Y_c)_{\overline{K}}$ with $Y_{\overline{K}}$ using its cocycle c. The point corresponding to q satisfies $c_\sigma \sigma_Y(y) = y$. Since K is henselian, its valuation extends uniquely to $\overline{K}$. The point y is defined over a finite extension, so properness of $\mathscr{Y}$ gives a reduction $\bar{y}$. Its image is the reduction of $\pi_c(q)$, which lies in $\mathscr{Z}$; hence $\bar{y}$ lies in $\mathscr{R}$ by (3.11).

Finite étaleness over the henselian valuation ring gives an equivariant reduction bijection [Stacks, Lemma 10.153.7, Tag 04GK]

$$R(\overline{K}) \xrightarrow{\sim} \mathscr{R}(\overline{\mathbf{F}}_v). \tag{3.13}$$

This bijection concerns only the finite ramification divisor. It commutes with Galois and the extended deck action, so reduction of $c_\sigma \sigma_Y(y) = y$ fixes $\bar{y}$ under the twisted action. Hence the unique lift of $\bar{y}$ is fixed by every $c_\sigma \sigma_Y$ and descends to $r \in Y_c(K)$ with $\pi_c(r) \in Z(K)$.

For the last assertion, if the image of a local point reduces outside $\mathscr{Z}$, all local boundary intersection multiplicities are zero, and a sufficiently small neighbourhood gives the required W. Otherwise use r above and put $p = \pi_c(r) \in Q_j(K)$. The closure $\mathscr{Q}_{j,\lambda}$ of p is the degree-one component in (3.12), even when Q_j is not rational over k. Choose a local parameter u at r and an integral local equation z for $\mathscr{Q}_{j,\lambda}$ near its reduction. Ramification gives

$$z \circ \pi_c = u^{e_j} h(u), \qquad h(0) \in K^\times.$$

For q' sufficiently close to r and off the ramification divisor,

$$n_v(\mathscr{Q}_{j,\lambda}, \pi_c(q')) = e_j \operatorname{ord}_v(u(q')) + \operatorname{ord}_v(h(0)) \longrightarrow +\infty. \tag{3.14}$$

All other local boundary intersection multiplicities are zero by (3.12). The factor $h(0)$ need not be a unit; its valuation is fixed. If $Q_j = P_i$ is an original boundary point, choose the displayed intersection multiplicity at least n_i; an auxiliary boundary point imposes no condition. Such q' exist because Y_c is a smooth curve with the K-point r. Lemma 2.1 now gives the required nonempty open W. □

3.4. **The global argument.**

Proof of Theorem 1.1. The case $D = 0$ is ordinary weak approximation. Otherwise we use the auxiliary divisor Δ and the cover of Proposition 3.1. Fix $(x_v)_v \in \prod_v \mathcal{C}_v(\mathbf{P}^1_{\mathcal{O}_S}, D)$ and finitely many approximation conditions. Choose a finite $T \supseteq S$ containing the prescribed approximation places and those dividing $|H|$, such that π extends over $\mathcal{O}_T$ to a finite

morphism $\mathscr{Y} \to \mathbf{P}^1_{\mathcal{O}_T}$ with $\mathscr{Y}$ smooth and proper, G extends to a finite étale group scheme acting on $\mathscr{Y}$ over the base, and $\mathscr{Z}, \mathscr{R}$ are finite étale divisors with $\pi^*\mathscr{Q}_j = e_j\mathscr{R}_j$. By Lemma 2.1, choose nonempty open sets $V_v \subset U_\Delta(k_v) \cap \mathcal{C}_v(\mathbf{P}^1_{\mathcal{O}_S}, D)$, $v \in T$, compatible with the prescribed approximation conditions. Choose $a \in U_\Delta(k)$ in all V_v. The pointed twist $\pi_a : Y_a \to \mathbf{P}^1$ has a point $q \in Y_a(k)$ above a, so $Y_a \simeq \mathbf{P}^1_k$.

Choose a finite $T' \supseteq T$ such that π_a extends to a finite morphism from a smooth proper $\mathcal{O}_{T'}$-model of Y_a to $\mathbf{P}^1_{\mathcal{O}_{T'}}$, and the closures $\mathscr{R}_{j,a}$ of $(\pi_a^{-1}(Q_j))_{\mathrm{red}}$ satisfy

$$\pi_a^*\mathscr{Q}_j = e_j\mathscr{R}_{j,a} \tag{3.15}$$

as effective Cartier divisors. At $v \in T$ take a neighbourhood of q mapping into V_v. At $v \in T' \setminus T$, Lemma 3.2 (for q) gives a nonempty open set mapping to $\mathcal{C}_v(\mathbf{P}^1_{\mathcal{O}_S}, D) \cap U_\Delta(k_v)$. Weak approximation on $Y_a \simeq \mathbf{P}^1_k$ gives a rational point p in all these open sets, with $t = \pi_a(p) \in U_\Delta(k)$.

The prescribed approximation conditions and the Campana conditions at $T' \setminus S$ hold by construction. For $v \notin T'$, properness and (3.15) give

$$n_v(\mathscr{Q}_j, t) = e_j n_v(\mathscr{R}_{j,a}, p) \in e_j\mathbf{Z}_{\geq 0}.$$

By (3.12), this is the unique possibly positive local-component intersection multiplicity over Q_j. If $Q_j = P_i$, every positive value is at least $e_j \geq n_i$. All other local-component intersection multiplicities vanish. Thus $t \in \mathcal{C}_S(\mathbf{P}^1_{\mathcal{O}_S}, D)$ gives the required approximation. □

4. Hyperplane arrangements

Let $N = n + 3$ and let $E_i = \{\ell_i = 0\}$ be as in Theorem 1.2. Put $U = \mathbf{P}^n \setminus \bigcup_i E_i$, and let $\mathscr{E}_i$ be the closure of E_i in $\mathbf{P}^n_{\mathcal{O}_S}$. Let A be the $N \times (n+1)$ matrix of the forms ℓ_i, and choose a basis of their two relations:

$$\sum_i b_i\ell_i = \sum_i c_i\ell_i = 0, \qquad B = \begin{pmatrix} b_1 & \cdots & b_N \\ c_1 & \cdots & c_N \end{pmatrix}. \tag{4.1}$$

General position is equivalent to $\Delta_{ij} := b_ic_j - c_ib_j \neq 0$ for $i \neq j$.

Let $T \supseteq S$ be a finite set containing the places above 2 such that outside T the matrices A and B are integral with unit maximal minors and the forms define the boundary closures $\mathscr{E}_i$. The unit-minor condition gives $\ker(B : \mathcal{O}_v^N \to \mathcal{O}_v^2) = A\mathcal{O}_v^{n+1}$, with AX primitive if and only if X is primitive. Each E_i is geometrically integral, so it has just one local irreducible component at every place. For $v \notin T$ and primitive $X \in \mathcal{O}_v^{n+1}$ off the boundary, the pulled-back ideal is generated by $\ell_i(X)$; hence its intersection multiplicity in the sense of [NS24, Definitions 2.3–2.4] is $n_v(\mathscr{E}_i, [X]) = \mathrm{ord}_v(\ell_i(X))$.

Put

$$Z = \bigcup_{i<j}(E_i \cap E_j), \qquad \mathscr{Z} = \bigcup_{i<j}(\mathscr{E}_i \cap \mathscr{E}_j), \qquad \mathcal{W} = \mathbf{P}^n_{\mathcal{O}_S} \setminus \mathscr{Z}.$$

Require also that $\mathcal{W}(\mathcal{O}_v) \neq \varnothing$ for every $v \notin T$. For $a \in U(k)$ and $v \notin T$, the condition $a \in \mathcal{W}(\mathcal{O}_v)$ says precisely that, after choosing a primitive local representative $\widetilde{a}$, at most one value $\ell_i(\widetilde{a})$ has positive valuation. Such a exists by strong approximation on $\mathbf{P}^n \setminus Z$.

For $a \in U(k)$, choose a homogeneous representative $\widetilde{a}$ and put $a_i = \ell_i(\widetilde{a})$. Define

$$Y_a: \quad Q_b = \sum_i b_i a_i y_i^2 = 0, \qquad Q_c = \sum_i c_i a_i y_i^2 = 0 \quad \subset \mathbf{P}^{n+2}_k. \tag{4.2}$$

The linear embedding $\iota(x) = [\ell_1(x) : \cdots : \ell_N(x)]$ identifies $\mathbf{P}^n$ with the subspace defined by (4.1).

Proposition 4.1. *The variety Y_a is a smooth geometrically integral intersection of two quadrics of dimension n. There is a finite morphism of degree 2^{n+2},*

$$\pi_a : Y_a \longrightarrow \mathbf{P}^n_k, \qquad \iota(\pi_a(y)) = [a_1 y_1^2 : \cdots : a_N y_N^2],$$

with $q_0 = (1 : \cdots : 1) \in Y_a(k)$ and $\pi_a(q_0) = a$. The morphism π_a does not depend on the choice of homogeneous representative $\widetilde{a}$.

Proof. Let $M_b = \operatorname{diag}(b_i a_i)_{i=1}^N$ and $M_c = \operatorname{diag}(c_i a_i)_{i=1}^N$. The discriminant of the pencil $sQ_b + tQ_c$ is

$$\det(sM_b + tM_c) = \left(\prod_{i=1}^N a_i\right) \prod_{i=1}^N (sb_i + tc_i).$$

Since $a \in U(k)$, all a_i are nonzero. Moreover, $\Delta_{ij} \neq 0$ for $i \neq j$, so the linear factors $sb_i + tc_i$ are nonzero and pairwise nonproportional. Thus the discriminant has N distinct zeros in $\mathbf{P}^1_{\overline{k}}$. By Reid's criterion [Rei72, Proposition 2.1], applied over $\overline{k}$, Y_a is a smooth complete intersection of two quadrics of dimension $N - 3 = n$.

The finite morphism

$$F_a : \mathbf{P}^{N-1} \longrightarrow \mathbf{P}^{N-1}, \qquad [y_1 : \cdots : y_N] \longmapsto [a_1 y_1^2 : \cdots : a_N y_N^2]$$

has $Y_a = F_a^{-1}(\iota(\mathbf{P}^n))$, and hence induces the stated finite morphism π_a. The generic fibre of $(\pi_a)_{\overline{k}}$ is obtained over $\overline{k}(\mathbf{P}^n)$ by adjoining square roots of

$$f_i = \frac{a_N \ell_i}{a_i \ell_N}, \qquad 1 \leq i < N.$$

These squareclasses are independent, since $\operatorname{ord}_{E_j}(f_i) = \delta_{ij}$ for $1 \leq i, j < N$. Thus the generic fibre of $(\pi_a)_{\overline{k}}$ is integral of degree 2^{N-1}. Every geometric irreducible component of Y_a has dimension n and therefore dominates $\mathbf{P}^n$ under the finite morphism π_a. Together with smoothness, this proves that Y_a is geometrically integral and that $\deg(\pi_a) = 2^{N-1} = 2^{n+2}$.

The relations (4.1) give $q_0 = (1 : \cdots : 1) \in Y_a(k)$ and $\pi_a(q_0) = a$. Replacing $\widetilde{a}$ by $\lambda\widetilde{a}$ multiplies all a_i by the same $\lambda \in k^\times$, leaving both Y_a and F_a unchanged. Thus π_a is independent of this choice. □

For $n \geq 3$, Proposition 2.2 gives ordinary weak approximation on every Y_a. For $n = 2$, we make a suitable choice of a.

Proposition 4.2. *Suppose $n = 2$. For any nonempty open sets $V_v \subset U(k_v)$, $v \in T$, there exists $a \in U(k)$ belonging to all V_v such that $\operatorname{Br}(Y_a) = \operatorname{Br}(k)$ and $a \in \mathcal{W}(\mathcal{O}_v)$ for every $v \notin T$. This Y_a satisfies ordinary weak approximation.*

Proof. The five singular members of the pencil are defined over k. The member $c_iQ_b - b_iQ_c = -\sum_{h\neq i}\Delta_{ih}a_hy_h^2$ has rank four and discriminant squareclass

$$\delta_i = d_i\prod_{h\neq i}a_h \in k^\times/k^{\times 2}, \qquad d_i = \prod_{h\neq i}\Delta_{ih}, \qquad \prod_{i=1}^{5}\delta_i = 1. \tag{4.3}$$

The product relation follows because each a_h occurs four times and $\prod_i d_i = \prod_{i<h}\Delta_{ih}^2$. By [Wit07, Theorem 3.37 and Example 3.41], it suffices to choose a so that $\delta_1,\dots,\delta_4$ are independent: then all five classes are nontrivial, and their product is their only nontrivial relation.

Choose four distinct finite places $w_1,\dots,w_4 \notin T$ with sufficiently large residue fields. At w_j, impose, using a primitive local representative of a,

$$\operatorname{ord}_{w_j}(a_j) = 1, \qquad \operatorname{ord}_{w_j}(a_i) = 0 \quad (i\neq j). \tag{4.4}$$

This defines a nonempty open subset of $U(k_{w_j})$: lift a residue point of E_j lying on none of the other lines, with intersection multiplicity exactly one along $\mathscr{E}_j$. The auxiliary open sets lie in $\mathcal{W}(\mathcal{O}_{w_j})$. Strong approximation on $\mathbf{P}^2\setminus Z$, applied to these open sets and the V_v, gives $a\in U(k)$ satisfying all the conditions and integral in $\mathcal{W}$ outside $T\cup\{w_1,\dots,w_4\}$. The prescribed intersection multiplicities at the w_j show that in fact $a\in\mathcal{W}(\mathcal{O}_v)$ for every $v\notin T$.

Since each d_i is a unit at the w_j, the valuation parities of $\delta_1,\dots,\delta_4$ at these places form the matrix

$$M = \begin{pmatrix} 0 & 1 & 1 & 1\\ 1 & 0 & 1 & 1\\ 1 & 1 & 0 & 1\\ 1 & 1 & 1 & 0 \end{pmatrix}, \qquad M^2 = I_4 \quad \text{over } \mathbf{F}_2.$$

These parities are independent of the choice of homogeneous representative: rescaling multiplies each discriminant representative by a fourth power. Thus a square relation $\prod_{i=1}^4\delta_i^{\epsilon_i}=1$ gives $M\epsilon = 0$, hence $\epsilon = 0$. Wittenberg's criterion yields $\operatorname{Br}(Y_a)/\operatorname{Br}(k) = 0$. Ordinary weak approximation now follows from Proposition 2.2. □

Lemma 4.3. *Let $a\in U(k)$ and let $v\notin T$ be finite. Suppose that the reduction of a lies outside $\mathscr{Z}$. Then there exists $y_v\in Y_a(k_v)$ with $\pi_a(y_v)\in U(k_v)\cap\mathcal{C}_v(\mathbf{P}^n_{\mathcal{O}_S},D)$.*

Proof. Put $R=\mathcal{O}_v$ and $\varpi=\varpi_v$, and rescale $\widetilde{a}$ locally to be integral and primitive. If all $a_i=\ell_i(\widetilde{a})$ are units, take $y_v=q_0$. Otherwise exactly one, say a_j, has positive valuation. Since ℓ_j is primitive over R, choose $d\in R^{n+1}$ with $\ell_j(d)=1$ and set

$$X = \widetilde{a} + (\varpi^2-1)a_jd.$$

Then $X\equiv\widetilde{a}\pmod{\varpi}$ is primitive, $\ell_j(X)=\varpi^2a_j$, and $\ell_i(X)/a_i\in 1+\varpi R$ for $i\neq j$. These latter ratios are squares by Hensel's lemma. Set $y_j=\varpi$ and choose $y_i\in 1+\varpi R$ with $y_i^2=\ell_i(X)/a_i$ for $i\neq j$. The relations among the ℓ_i give $y_v\in Y_a(k_v)$ and $\pi_a(y_v)=[X]$.

Consequently

$$n_v(\mathscr{E}_i,[X])=\begin{cases}\operatorname{ord}_v(a_j)+2\geq 3, & i=j,\\ 0, & i\neq j.\end{cases} \tag{4.5}$$

All $\ell_i(X)$ are nonzero, so $[X]\in U(k_v)$, as required. □

Proof of Theorem 1.2. Fix $(x_v)_v\in\prod_v \mathcal{C}_v(\mathbf{P}^n_{\mathcal{O}_S},D)$ and finitely many approximation conditions. Choose a finite $T\supseteq S$ containing the prescribed approximation places and those above 2, such that, for every $v\notin T$, the matrices A,B are integral with unit maximal minors, the forms ℓ_i define the boundary closures $\mathscr{E}_i$, and $\mathcal{W}(\mathcal{O}_v)\neq\varnothing$. By Lemma 2.1, choose nonempty open sets $V_v\subset U(k_v)\cap\mathcal{C}_v(\mathbf{P}^n_{\mathcal{O}_S},D)$, $v\in T$, compatible with the prescribed approximation conditions. For $n\geq 3$, strong approximation on $\mathbf{P}^n\setminus Z$ gives $a\in U(k)$ in all V_v with $a\in\mathcal{W}(\mathcal{O}_v)$ for every $v\notin T$. For $n=2$, use Proposition 4.2, which also ensures $\operatorname{Br}(Y_a)=\operatorname{Br}(k)$. In both cases, Y_a satisfies ordinary weak approximation.

Let $T'\supseteq T$ be finite and contain every place where a chosen global coefficient a_i is not a unit. At $v\in T$, use a neighbourhood of q_0 mapping into V_v. At $v\in T'\setminus T$, Lemma 4.3 and Lemma 2.1 give a nonempty open set whose image lies in $U(k_v)\cap\mathcal{C}_v(\mathbf{P}^n_{\mathcal{O}_S},D)$. This includes the four auxiliary places when $n=2$. Weak approximation on Y_a gives $y\in Y_a(k)$ in all these open sets; we may require every y_i to be nonzero.

Put $x=\pi_a(y)$. At $v\notin T'$, take primitive integral coordinates for y. As all a_i are units, $w=(a_iy_i^2)_i$ is primitive and belongs to $\ker B$, so $w=AX$ with X integral and primitive. Therefore

$$n_v(\mathscr{E}_i,x)=\operatorname{ord}_v(a_iy_i^2)=2\operatorname{ord}_v(y_i)\in 2\mathbf{Z}_{\geq 0}. \tag{4.6}$$

These are precisely the local-component intersection multiplicities. The Campana conditions at $T\setminus S$ hold because $x\in V_v$, and those at $T'\setminus T$ hold by local repair. Thus $x\in\mathcal{C}_S(\mathbf{P}^n_{\mathcal{O}_S},D)$ gives the required approximation. □

5. Two transverse quadrics

Let C_1,C_2 be as in Theorem 1.3, with equations $q_i=0$, and let $U=\mathbf{P}^n\setminus(C_1\cup C_2)$. Let $\mathscr{C}_i$ be the closure of C_i in $\mathbf{P}^n_{\mathcal{O}_S}$. For $a\in U(k)$ choose $\widetilde{a}=(a_0,\dots,a_n)$, put $c_i=q_i(\widetilde{a})$, and define

$$Y_a:\quad q_1(X_0,\dots,X_n)=c_1Z_1^2,\qquad q_2(X_0,\dots,X_n)=c_2Z_2^2\quad\subset\mathbf{P}^{n+2}_k. \tag{5.1}$$

The projection gives $\pi_a:Y_a\to\mathbf{P}^n_k$ with the rational point $P_0=[a_0:\cdots:a_n:1:1]$ above a.

Proposition 5.1. *The variety Y_a is a smooth geometrically integral intersection of two quadrics of dimension n, and π_a is finite flat of degree four.*

Proof. The X-coordinates cannot all vanish. On each affine chart of the base, adjoining the two roots gives a free algebra with basis $1,z_1,z_2,z_1z_2$. Hence π_a is finite flat of degree 4. For any linear form L, the squareclasses of q_i/L^2 over $\overline{k}(\mathbf{P}^n)$ are independent; this gives geometric integrality. A relation between the gradients has the form

$$\lambda(\nabla q_1,-2c_1Z_1,0)+\mu(\nabla q_2,0,-2c_2Z_2)=0.$$

If both Z_i are nonzero, then $\lambda = \mu = 0$. If precisely one vanishes, smoothness of the corresponding quadric gives the same conclusion. If both vanish, transversality gives independence of the gradients. Thus Y_a is smooth. □

In dimension two, transversality says that the conics meet in four distinct geometric points. Adjunction gives $-K_{Y_a} = (n-1)\mathcal{O}_{Y_a}(1)$, so the surface cover is a del Pezzo surface of degree 4. The degree of the map π_a is 4 in every dimension, whereas the hyperplane cover has degree 2^{n+2}; the covers in both cases are smooth complete intersections of two quadrics in their projective embeddings.

Let $T \supseteq S$ be a finite set containing the places above 2 such that outside T the q_i are integral and define the boundary closures, the quadrics have smooth reduction, and their intersection remains transverse. Since $n \geq 2$, each smooth quadric C_i is geometrically integral: a reducible quadratic form over $\bar{k}$ would define two hyperplanes (or a double hyperplane), and hence a singular hypersurface. Thus C_i has just one local irreducible component at every place. For $v \notin T$ and primitive integral coordinates X, the pulled-back ideal of $\mathscr{C}_i$ is generated by $q_i(X)$, so

$$n_v(\mathscr{C}_i, [X]) = \operatorname{ord}_v(q_i(X)) \qquad (v \notin T)$$

is the intersection multiplicity of [NS24, Definitions 2.3–2.4]. Put

$$Z = C_1 \cap C_2, \qquad \mathscr{Z} = \mathscr{C}_1 \cap \mathscr{C}_2, \qquad \mathcal{W} = \mathbf{P}^n_{\mathcal{O}_S} \setminus \mathscr{Z}.$$

Require also that $\mathcal{W}(\mathcal{O}_v) \neq \varnothing$ for every $v \notin T$. Since Z has codimension two, strong approximation on $\mathbf{P}^n \setminus Z$ lets us choose a whose reduction belongs to at most one boundary quadric outside the finite approximation set.

Lemma 5.2. *Let $a \in U(k)$ and let $v \notin T$ be finite. Suppose that the reduction of a lies outside $\mathscr{Z}$. Then there exists $P_v \in Y_a(k_v)$ such that*

$$\pi_a(P_v) \in U(k_v) \cap \mathcal{C}_v(\mathbf{P}^n_{\mathcal{O}_S}, D), \qquad \alpha(P_v) = \alpha(P_0) \quad (\alpha \in \operatorname{Br}(Y_a)).$$

Proof. Put $R = \mathcal{O}_v$ and $\varpi = \varpi_v$. Rescale $\widetilde{a}$ to be primitive and work in an affine chart with one unit coordinate equal to 1. Rescaling by λ replaces c_i by $\lambda^2 c_i$; the isomorphism $[X : Z_1 : Z_2] \mapsto [\lambda X : Z_1 : Z_2]$ preserves the projection and the distinguished point. Thus the local equations are

$$f_1(x) = c_1 z_1^2, \qquad f_2(x) = c_2 z_2^2, \qquad c_i = f_i(a) \in R \setminus \{0\}, \qquad P_0 = (a, 1, 1). \tag{5.2}$$

If both c_i are units, take $P_v = P_0$. Otherwise, after interchanging the indices, we may assume $\operatorname{ord}_v(c_1) > 0$ and $c_2 \in R^\times$. Smoothness of the reduction of C_1 gives a coordinate x_r with $\partial \bar{f}_1/\partial x_r(\bar{a}) \neq 0$. Fixing the other base coordinates, Hensel's lemma gives $x \equiv a \pmod{\varpi}$ with $f_1(x) = c_1 \varpi^2$. Since $f_2(x)/c_2 \in 1 + \varpi R$, choose $z_2 \in 1 + \varpi R$ with $z_2^2 = f_2(x)/c_2$ and put $P_v = (x, \varpi, z_2)$. Then

$$n_v(\mathscr{C}_1, \pi_a(P_v)) = \operatorname{ord}_v(c_1) + 2 \geq 3, \qquad n_v(\mathscr{C}_2, \pi_a(P_v)) = 0. \tag{5.3}$$

Both quadratic values are nonzero, so the image belongs to $U(k_v)$.

For $n \geq 3$, the Brauer assertion follows from Proposition 2.2. For $n = 2$, let $\mathcal{M}$ be the relative smooth locus of (5.2). The line

$$A = \{x = \bar{a},\ z_2 = 1\} \simeq \mathbf{A}^1_{\mathbf{F}_v}$$

lies in $\mathcal{M}_{\mathbf{F}_v}$: along A, the Jacobian minor in x_r, z_2 equals $-2\bar{c}_2\,\partial\bar{f}_1/\partial x_r(\bar{a}) \neq 0$. Both P_0 and P_v are sections of $\mathcal{M}$ reducing into A, with z_1 respectively equal to 1 and 0 on reduction. For $\alpha \in \mathrm{Br}(Y_a)$, the normalized class $\beta = \alpha - \alpha(P_0)$ is 2-torsion by Proposition 2.2 and evaluation at P_0. Restricting β to $\mathcal{M}_{k_v}$ and applying Lemma 2.3 gives $\beta(P_v) = \beta(P_0) = 0$, which proves the assertion. □

Proof of Theorem 1.3. Fix $(x_v)_v \in \prod_v \mathcal{C}_v(\mathbf{P}^n_{\mathcal{O}_S}, D)$ and finitely many approximation conditions. Choose a finite $T \supseteq S$ containing the prescribed approximation places and those above 2, such that, for every $v \notin T$, the forms q_i are integral and define the boundary closures $\mathscr{C}_i$, the reductions of C_1, C_2 are smooth and meet transversely, and $\mathcal{W}(\mathcal{O}_v) \neq \varnothing$. By Lemma 2.1, choose nonempty open sets $V_v \subset U(k_v) \cap \mathcal{C}_v(\mathbf{P}^n_{\mathcal{O}_S}, D)$, $v \in T$, compatible with the prescribed approximation conditions. Strong approximation on $\mathbf{P}^n \setminus Z$ gives $a \in U(k)$ belonging to all V_v and satisfying $a \in \mathcal{W}(\mathcal{O}_v)$ for every $v \notin T$. Construct Y_a and P_0, and choose a finite $T' \supseteq T$ containing the places where either global coefficient c_i is not a unit.

For $v \in T' \setminus T$, choose P_v by Lemma 5.2. Put $Q_v = P_v$ at these places and $Q_v = P_0$ elsewhere. The lemma and reciprocity give

$$\sum_v \mathrm{inv}_v\,\alpha(Q_v) = \sum_v \mathrm{inv}_v\,\alpha(P_0) = 0 \qquad (\alpha \in \mathrm{Br}(Y_a)).$$

Here $\alpha(P_0) \in \mathrm{Br}(k)$ is one fixed global class, and each local modification preserves its evaluation. In particular, we do not assume that the surface has trivial Brauer group. Thus $(Q_v)_v \in Y_a(\mathbf{A}_k)^{\mathrm{Br}(Y_a)}$, and Proposition 2.2 gives $P \in Y_a(k)$ sufficiently close to Q_v for every $v \in T'$. Require $\pi_a(P) \in V_v$ at T, and a Campana image at $T' \setminus T$. By openness, we may also require $Z_1 Z_2 \neq 0$.

Set $x = \pi_a(P)$. At $v \notin T'$, take primitive integral coordinates $[X_0 : \cdots : X_n : Z_1 : Z_2]$ for P. The vector X is primitive: otherwise $q_i(X) \in \mathfrak{m}_v^2$, and, since c_i are units, the equations would force both Z_i into $\mathfrak{m}_v$, contradicting primitivity of the full coordinate vector. Hence

$$n_v(\mathscr{C}_i, x) = \mathrm{ord}_v(q_i(X)) = 2\,\mathrm{ord}_v(Z_i) \in 2\mathbf{Z}_{\geq 0}. \tag{5.4}$$

These are the local-component intersection multiplicities. The Campana conditions at $T \setminus S$ hold because $x \in V_v$, and those at $T' \setminus T$ hold by local repair. Thus $x \in \mathcal{C}_S(\mathbf{P}^n_{\mathcal{O}_S}, D)$ gives the required approximation. □

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